Conjecture 00000002571: the rank-one boundary of the printed bound

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1 The exact inequality as printed

The conjecture asserts the explicit bound

$$D(f) \le 16\big(\log \operatorname{rank}(M_f)\big)^2.$$

There is no restriction to rank at least two, additive constant, or replacement of the rank by its maximum with two. We refute this exact statement at rank one. This does not refute the usual asymptotic log-rank conjecture or a version with a low-rank regularization, and does not address the separate optimal-exponent assertion.

There are two common matrix encodings; we give a witness for each. For the $\{0,1\}$ encoding, take $f(a,b) = a \wedge b$. For the sign encoding $M_f(a,b) = (-1)^{f(a,b)}$, take $f(a,b) = a \oplus b$. Ordering the inputs as false, true gives

$$M_{\wedge} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad M_{\oplus}^{\mathrm{sign}} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

Both real matrices have rank one. Each is a nonzero outer product, so its rank is at most one, and a nonzero 1×1 minor proves the reverse inequality. This is the actual matrix rank computed in the formal proof, not a separately assigned rank value.

2 A positive communication cost

Suppose Bob must output the answer and no communication occurs. His answer can depend on his own input b, but cannot depend on Alice's private input a. XOR and AND both violate this condition:

$$(0 \oplus 0) \neq (1 \oplus 0), \qquad (0 \wedge 1) \neq (1 \wedge 1).$$

They also depend on Bob's input, so choosing Alice as the sole output party does not permit zero communication either. If both parties must output the answer, zero communication is likewise impossible.

For the Bob-output convention, exactly one bit suffices: Alice sends her input and Bob evaluates $f(a,b)$. Thus $D(f) = 1$ for each witness in that convention. We use only the positive lower bound for the contradiction, so no exact one-bit claim is needed for the convention requiring both parties to output.

3 The formal protocol and minimum

The Lean inductive type `Protocol` has Alice and Bob message nodes and Bob-output leaves. At an Alice node the transmitted bit is a function of Alice's local input; at a Bob node it is a function of Bob's local input. The chosen subtree incorporates the transcript. A leaf may output any function of Bob's local input. In particular, the model does not incorrectly force a zero-message protocol to output a global constant.

The interpreter evaluates the actual protocol on a pair of inputs. Cost is worst-case tree depth. Correctness quantifies over all input pairs. An explicit one-bit correct protocol establishes a nonempty set of achievable costs, and `D` is its least natural-number cost, defined with `Nat.find`. The proof of `zero_cost_local` classifies every zero-cost protocol. It then proves the lower bounds and matches them with the explicit upper bounds; the number one is not inserted as a definition of communication complexity.

Theorems `rank_bit_and` and `rank_sign_xor` use Mathlib's `Matrix.rank` over $\mathbf{R}$, outer-product rank bounds and the rank of a 1×1 submatrix. The final two counterexample theorems combine these ranks with the actual minimum costs and $\log_2 1 = 0$. The base of the logarithm does not affect this rank-one obstruction.

The standard deterministic communication model and real $\{0,1\}$-matrix convention are described in Ben-Sasson, Lovett and Zewi, *An additive combinatorics approach to the log-rank conjecture*, section 1.1. Our contradiction concerns the exact low-rank formula printed in this competition, not the asymptotic notation used in that paper.

4 Verification and scope

The accompanying project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

The build, transitive axiom audit, official fresh kernel replay and source hashes are recorded in the package. No `sorry`, `native_decide`, or additional axiom is used. The original bilingual statement is preserved in `SOURCE.md`. OpenAI Codex assisted the proof and report; a separate supervising Codex agent reviewed the formal chain. This is AI review, not human certification. Organizer acceptance is pending.